Analyzing Biodiversity and Wildfire Outcomes on Indigenous and Non-Indigenous Lands Across Oregon

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Abstract

As catastrophic wildfires increasingly reshape the American West, traditional suppression methods are no longer enough to protect vulnerable ecosystems. We investigated the relationships between wildfire severity, prescribed burns, federally recognized Indigenous land, and the presence of fire resistant and keystone plant species in Oregon. To facilitate our analysis, we constructed a grid covering Oregon to aggregate our data on a spatial level to maximize accuracy while minimizing computational burden. Our results indicate that prescribed burns impact fire duration and size, with cells that had prescribed burns having more small fires and more large fires (perhaps because areas prone to large wildfires are targets for prescribed burns).

Introduction & Motivation

Problem Wildfires are becoming increasingly frequent and severe across Oregon due to climate change and decades of fire-exclusion policies that have altered forest structure and increased landscape fuel loads. Indigenous communities have stewarded Pacific Northwest forests for thousands of years through practices such as cultural burning, but colonial policies disrupted these stewardship systems, contributing to declining forest resilience and heightened wildfire risk. As Oregon's mature and old-growth forests – and the communities that depend on them – face growing threats, there is increasing interest in understanding how Indigenous land stewardship may contribute to healthier forests and reduced wildfire impacts.

Research Question: This project examines whether federally recognized Indigenous lands in Oregon exhibit different forest composition and wildfire severity outcomes than nearby ecologically similar non-Indigenous lands. Specifically, we investigate whether Indigenous lands are associated with greater forest compositional diversity, lower wildfire severity after accounting for environmental factors, and which ecological characteristics are most predictive of wildfire severity.

Impact & stakeholders: Findings from this research can support Indigenous community leaders, forest management agencies, fire prevention officials, emergency preparedness organizations, first responders, and Oregon communities vulnerable to wildfire. By providing evidence on relationships between forest composition, stewardship practices, and wildfire outcomes, the project aims to inform collaborative fire management strategies that strengthen ecological resilience while supporting more effective wildfire risk reduction.

Background & Related Work

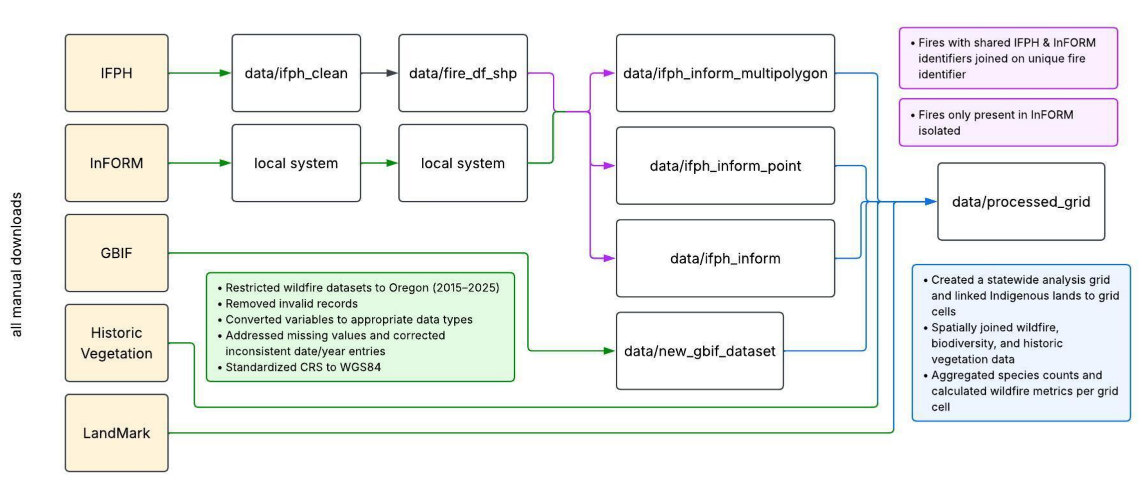
While existing work has advanced understanding of Indigenous stewardship [1][2], wildfire ecology [1][2][3], and forest resilience [3], our project addresses a quantitative gap by integrating geospatial wildfire, forest composition, biodiversity, and Indigenous land datasets to evaluate landscape-scale relationships across Oregon. Rather than focusing primarily on historical documentation [3], policy recommendations [2], or ecological theory [4], we apply a reproducible data science framework to compare federally recognized Indigenous lands with nearby ecologically similar non-Indigenous lands. This approach allows us to assess whether differences in forest diversity and ecological characteristics are associated with differences in wildfire severity outcomes while recognizing the observational nature of these relationships and the importance of Indigenous expertise in interpreting results.

Data, Engineering & Ethics

Sources & scale: This project integrates publicly available ecological and geospatial datasets from LandMark, GBIF, NIFC InFORM, NIFC Interagency Wildland Fire Perimeter History (IFPH), and Oregon Historic Vegetation. Dataset sizes range from 4,262 to over 1.2 million records, with total storage requirements ranging from approximately 14 MB to 2 GB. Data are distributed under varying open-access licenses, including CC BY-SA 4.0 (LandMark), Creative Commons licenses (GBIF), CC0 1.0 (Historic Vegetation), and agency-specific access terms for NIFC datasets; all applicable attribution and responsible-use requirements are followed. The full dataset documentation, licensing details, and preprocessing notes are provided in the project's M2 Data Summary.

Ethics & responsible use: This project uses only publicly available ecological and geospatial datasets and contains no personally identifiable information (PII) or human subjects data. To protect Indigenous communities, we recognize that public datasets cannot fully represent Indigenous knowledge or stewardship practices and interpret results through the CARE (Collective Benefit, Authority to Control, Responsibility, Ethics) principles. Analyses are framed as observational, avoiding unsupported causal claims or generalizations across Indigenous nations. We report landscape-level ecological patterns rather than cultural conclusions and avoid deficit-based comparisons of land management systems. Dataset-specific licensing and attribution requirements are followed, sensitive metadata (e.g., observer names and tribal identifiers) are excluded from analysis and redistribution where appropriate, and findings are to be presented to support collaborative wildfire stewardship while respecting Indigenous rights and data sovereignty.

Pipeline:

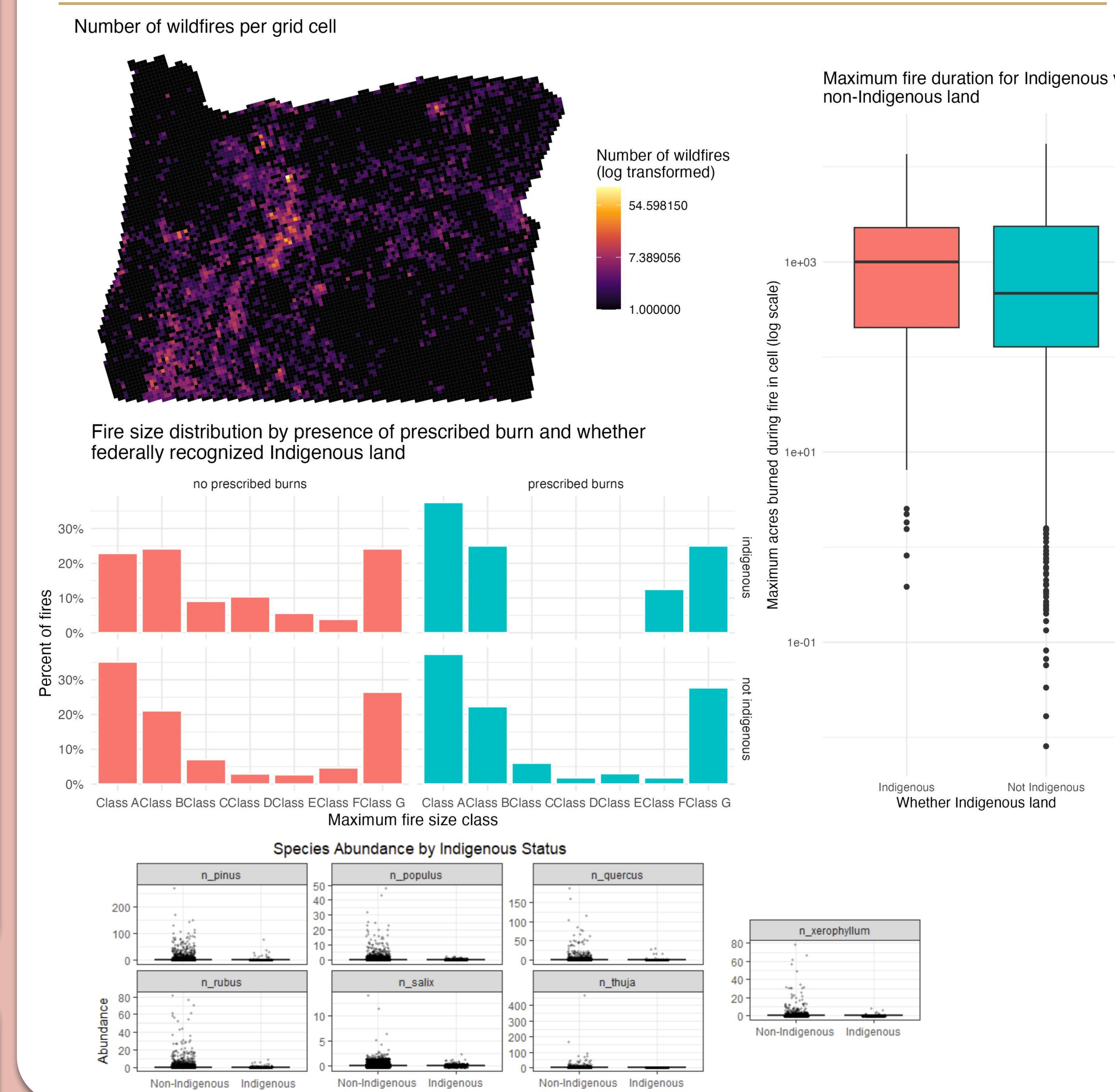


Methods & Evaluation

Approach: We consolidated and aggregated our data based on a constructed grid, with each cell being 5 by 5 km (25 square km, approx. 6178 acres). We used an equidistant coordinate system and the sf package to create the grid and then transformed it into WGS 84 coordinate system. We checked the presence of spatial data in each cell using intersection for multipolygons and containment for points - this is the most inclusive method, where a multipolygon that has any intersection with a cell is included in that cell aggregations. Multipolygon fire perimeters that span several cells will be counted for each cell they appear in; for our modeling, it creates the most accurate record of the fire history of each cell. We used a combination of statistical inference, regression modeling, and machine learning to evaluate relationships between Indigenous lands, forest composition, biodiversity, and wildfire severity. Differences in forest compositional diversity between Indigenous and non-Indigenous lands were assessed using independent t-tests (or Mann–Whitney U tests when assumptions were violated) and multiple linear regression predicting Shannon diversity. Wildfire severity (burned area, fire duration, and fire management complexity) was modeled using multiple linear regression, Gamma generalized linear models, and ordinal logistic regression, while Random Forest and XGBoost identified the ecological variables most predictive of wildfire severity.

Validation: Models were compared against baseline models (plant abundance or vegetation-only predictors), validated using 5-fold cross-validation and residual diagnostics, and evaluated using p-values, Cohen's d, R^2, RMSE, MAE, and AIC. Random Forest and XGBoost additionally provided variable importance, SHAP values, and partial dependence plots to improve model interpretation.

Results & Visualizations



Discussion, Limitations & Conclusions

Discussion: Wildfire patterns were broadly similar across federally recognized Indigenous and non-Indigenous lands. Prescribed burns were associated with a higher proportion of small fires, although the frequency of the largest fires remained comparable. Higher observed plant abundances on non-Indigenous lands likely reflect differences in land area and sampling effort rather than true ecological differences. As an observational study, these findings indicate associations rather than causal relationships.

Limitations & future work: Biodiversity metrics should be interpreted as proxies rather than comprehensive measures of Oregon's biodiversity. Analyses were limited to seven focal species, and GBIF observations are concentrated in more accessible areas, creating geographic sampling bias. Future studies should incorporate standardized biodiversity surveys, additional diversity indices (e.g., Shannon diversity), and environmental variables such as climate, vegetation type, and topography.

Conclusions: This study demonstrates the value of integrating publicly available biodiversity and wildfire datasets to examine landscape-scale ecological patterns while highlighting the importance of careful interpretation. Future research using more comprehensive biodiversity data and environmental controls will improve understanding of how Indigenous stewardship and prescribed fire relate to biodiversity conservation and wildfire resilience.

Reproducibility & References

Repository: [https://github.com/Fiden-Supancich-McCord/data510-fire] · Key dependencies: [ ]

[1] Coughlan M. R., Johnston J. D., Derr K. M., Lewis D. G. and Johnson B. R. (2024). Pre-contact Indigenous fire stewardship: a research framework and application to a Pacific Northwest temperate rainforest. Front. Environ. Archaeol. 3:1347571. doi: 10.3389/fearc.2024.1347571.

[2] Coughlan, M., Serio, N., Loeb, H., Lewis, D., & Thompson, S. (n.d.). Indigenous fire stewardship for fire management and ecological restoration in the Pacific Northwest Ecosystem Workforce Program. https://www.nwfirescience.org/sites/default/files/publications/SynthesisReport_IFS_Full_Accessible_oct30.pdf

[3] Lundeberg, S. (2026). Study: Old-growth forests face highest wildfire risk in areas that once burned frequently. Philomath News, Oregon State University. https://philomathnews.com/study-old-growth-forests-face-highest-wildfire-risk-in-areas-that-once-burned-frequently/.

[4]Tallamy, D. (2024). Keystone native plants: marine West coast forests – ecoregion 7. National Wildlife Federation. https://www.nwf.org/-/media/Documents/PDFs/Garden-for-Wildlife/Keystone-Plants/NWF-GFW-keystone-plant-list-ecoregion-7-marine-west-coast-forest.pdf.